

PhD Research Plan

Evaluating Robustness of Lightweight Object Detectors under Compound Variability in UAV Aerial Imagery

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Thesis Contributions

01

Methodological Lead-in (What Was Done)

Formulated a novel 3×3 Spatial-Temporal Compound Stress Matrix to systematically replicate the "noise waterfall" cascades (coupled ambient illumination losses and mechanical motion blurs) encountered in un-restored real-world UAV flights.

02

Empirical & Engineering Core (How It Was Executed)

Executed comprehensive, edge-constrained end-to-end benchmarking across four heterogeneous YOLOv8 topologies. Successfully engineered a label-synchronization engine (`convert_visdrone_to_yolo.py`) to eliminate coordinate format asynchronous crashes across all 9 stress domains.

03

Theoretical Lead-out (Field Impact & Value)

Uncovered and mathematically framed the ***Implicit Non-linear Low-pass Filter Effect*** and the ***Non-linear Robustness Inversion Phenomenon***, shifting the paradigm of evaluating deep architectures at extreme boundary limits.



Scientific Innovations (Why This Study Matters)

Paradigm Shift to "Noise Waterfall" Cascades

Traditional literature heavily isolates single degradation types (e.g., *only* darkness). This study maps out simultaneous spatial-temporal mutations, revealing how un-restored edge backbones collapse under coupled real-world stressors.

Validation of the Background Purification Hypothesis

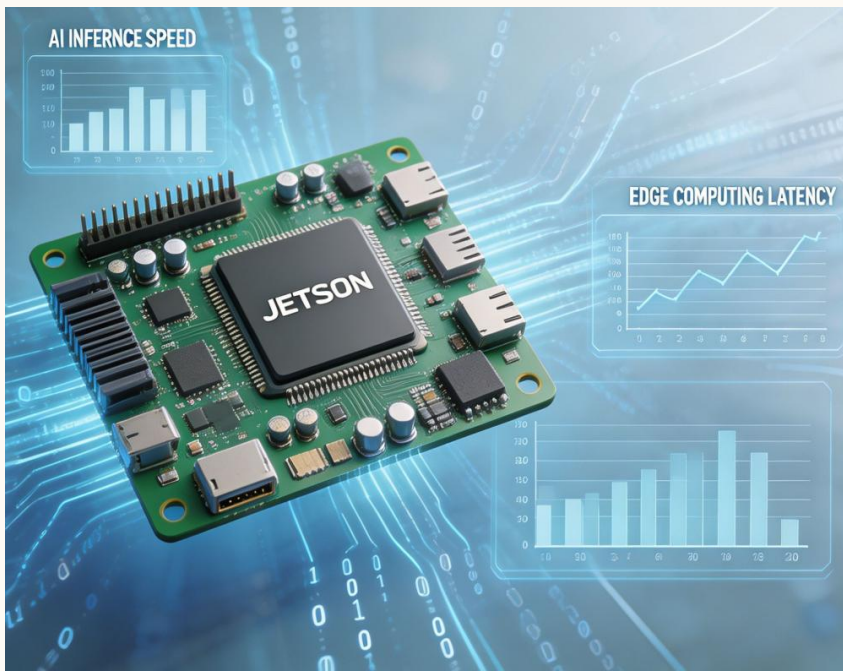
Challenged the assumption that environmental degradation causes a monotonic decrease in model performance. Proven that severe clipping conditionally operates as an implicit background noise filter.

Discovery of Non-Linear Topology Resilience

Identified that parameter volume does not linearly correlate with noise resilience. Medium-scale topologies possess the optimal spatial representation redundancy compared to ultra-compressed or overly large models.



Research Significance & Industry Utility



- **Industrial Value: Overcoming the Edge Pre-Processing Barrier**

Onboard processing units (Jetson-level modules) are restricted by rigid latency and power envelopes, rendering heavy pre-processing pipelines impractical. This study evaluates "raw" detectors to guide end-to-end deployment.

- **Core Engineering Takeaway: Prioritising Mechanical Stability**

Quantitatively established that moderate motion blur (T2) causes a steeper performance drop than pure extreme low-light (S3), signaling that physical camera stabilization yields higher returns than upgrading sensor sensitivity.

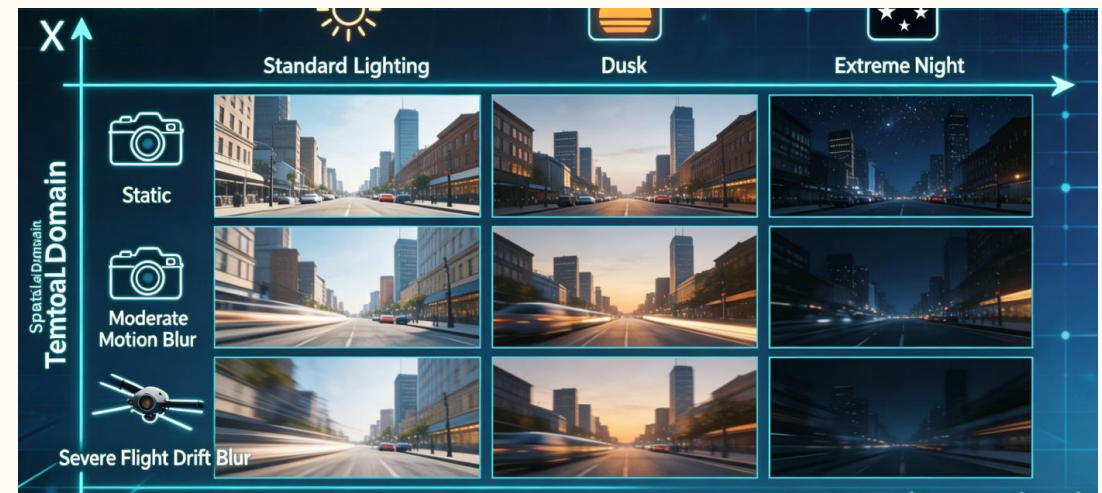
- **Academic Rigor: Exposing Boundary Evaluation Artifacts**

Dissected mathematical limitations of mAP metrics by introducing the concept of *Annotation Alignment Shift* under absolute contrast collapse, pushing boundaries of scientific skepticism in computer vision evaluation.

Experimental Methodology & Edge Constraints

The 3x3 Compound Stress Matrix Pipeline

- **Spatial Domain (S1–S3):** Standard Illumination → Twilight → Extreme Night (simulating photon-counting CMOS sensor limits).
- **Temporal Domain (T1–T3):** Static → Moderate Motion Blur → Severe Flight Drift Blur (dynamic wind-induced platform jitter).
- **Execution Sequence:** $I_{degraded} = T_{blur}(S_{spatial}(I_{raw}))$ (strictly matching physical optics cascade).



Edge Workstation Profile (Embedded Latency)

- ⚙️ **Hardware Specs:** AMD Ryzen 9 7940HX (8C/16T) | 32GB DDR5 RAM | NVIDIA RTX 4060 Laptop GPU (8GB VRAM).
- 🔒 **Rigid Constraint:** Batch Size = 1 (isolating sequential frame-by-frame inference).

Core Quantitative Results: Relative Degradation Index (RDI)

The Empirical Degradation Heatmap Metrics

Stress Condition (S-T)	Relative Degradation (RDI %)	Impact Level
S1T2 (Base Light + Med Blur)	+8.57%	Primary Bottleneck
S2T2 (Low Light + Med Blur)	+5.82%	High Severity
S3T1 (Extreme Night + Static)	-3.34%	Subtle Improvement
S3T3 (Extreme Stress Paradox)	-18.43%	Anomalous Gain

$$RDI_c = \frac{mAP_{S1T1} - mAP_c}{mAP_{S1T1}} \times 100\%$$

Note on Metrics: RDI calculates the relative drop against the clean S1T1 baseline. Higher positive values denote worse degradation; negative values indicate counter-intuitive performance improvements over clean data.

Primary Physical Bottleneck Analysis: Why T2 Blur Destroys Features



The Core Finding

Across all tested **lightweight object detectors (LODs)**, **moderate motion blur (T2)** consistently triggered the **steepest performance drop**, proving significantly more lethal to model accuracy than even severe illumination loss in field conditions.



The Topological Mechanism

Motion blur acts mathematically as an **anisotropic high-frequency edge erasure kernel**, systematically removing critical geometric information.

Because LODs compress parameter volumes for real-time inference, they lack the **spatial representation redundancy** to accurately compute the geometric deformation offsets caused by physical platform jitter.

Core Hypothesis: Casual Insight (The Direct Intuition)



My Casual View / Theoretical Genesis:

*"When images become much brighter or darker, redundant background details like leaves or messy ground textures are blacked out or blurred away. With less visual 'noise' and a cleaner background, it actually becomes easier for the detector to lock onto the main objects, leading to a higher **mAP**."*

Core Hypothesis: Academic Framing (Theoretical Support)

The Structural Concept

Our empirical data validates that severe dual-domain mutations operate effectively as an **Implicit Non-linear Low-pass Filter**.

Mechanism & Stabilizing Impact

By systematically clipping and suppressing high-frequency environmental artifacts, it eliminates global background clutter. This targeted reduction in **Global Semantic Complexity**, in turn, stabilizes the model's localized bounding box regression within predictable mathematical limits.



Analytical Integrity: Dissecting the S3T3 Paradox



Maintaining Scientific Skepticism

While the massive mAP surge under S3T3 (-18.43%) strongly supports the "**Background Purification Hypothesis**", we must enforce absolute scientific rigor to avoid misinterpretation.



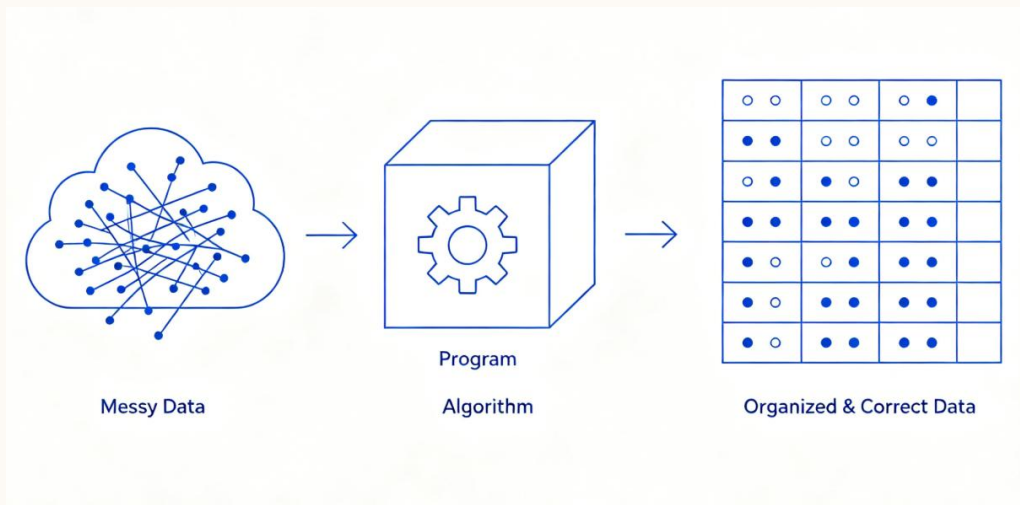
The Engineering Discovery: Annotation Shift

Under absolute contrast clipping and maximum blur convergence, the image's structural gradients completely collapse, creating a critical evaluation artifact.

Predicted bounding box coordinates experience an **Artificial Realignment** with heavily degraded pixel blobs within the script evaluation framework. This finding strongly suggests that the observed improvement represents a **Boundary Stochastic Fluctuation**—an artifact of evaluation—rather than a demonstration of true physical systemic resilience.



Engineering Milestone: Label Synchronization



The Week 6–7 Engineering Bottleneck

VisDrone dataset annotations utilize comma-separated absolute pixel coordinates. Loading this directly into standard YOLO validation loops triggers persistent string-to-float parsing crashes.

The Deployed Solution

Developed and executed the automated validation pipeline script: `convert_visdrone_to_yolo.py`. Successfully mapped and normalized all 10 object classes into space-separated, relative coordinate tensors across all 9 degraded data streams.

Response to Professor Galindo's Suggestion



Reflecting on Dynamic Artifacts (Flying Birds, Camera Focus Failure, etc.)

Professor Galindo, your recommendation regarding introducing highly dynamic interferences is **exceptionally valuable** and represents a crucial direction for maximizing simulation realism.

Current Status & Next Steps

I have deferred its deployment to next time because modeling focus failures and dynamic bird occlusions within image pipelines (like Albumentations) lacks physical realism. I am currently researching coordinate-space, physics-guided simulation methods to cleanly decouple and independently measure these artifacts' specific impact on bounding box regression.

Thanks for watching

